

Exercise 6

1) $\Gamma_m = 1 \cdot 10^6$ 7

Vi har at den relativt hastighets mellom systemet er v_A

$$v_A = 0.177024$$

$$L_N = 5.39693 \cdot 10^{41}$$

Vi har at

$$v' = \frac{v - v_{rel}}{1 - v_{rel}v}$$

som gir oss

$$v'_B = \frac{v_B - 2v_A}{1 - v_B(v_A)}$$

$$= \frac{-v_A - 2v_A}{1 + v_A^2} = \frac{-2v_A}{1 + v_A^2}$$

2)

Der har valgt Frame 1

$$P_\mu(A) = \gamma_A(m, mv_A, 0, 0)$$

$$P_\mu(B) = \gamma_A(m, -mv_A, 0, 0)$$

$$(\gamma_A = \gamma_B \text{ fordi } |v_A| = |v_B|)$$

3)

$$P'_\mu(A) = c_{\mu\nu} P_\nu(A)$$

$$\downarrow$$

$$E'_A = \gamma_{rel} E_A - v_{rel} \gamma_{rel} P_A$$

$$P'_A = \gamma_{rel} P_A - v_{rel} \gamma_{rel} E_A$$

Vi har fra tidligere at $v_{rel} = v_A$,

som gir oss $\gamma_{rel} = \gamma_A$ vi får:

$$E'_A = \gamma_A(\gamma_A m) - v_A \gamma_A(\gamma_A m v_A)$$

$$= \gamma_A^2 m - v_A^2 \gamma_A^2 m$$

$$= m(\gamma_A^2 - \gamma_A^2 v_A^2)$$

$$= m\left(\frac{1}{1-v_A^2} - \frac{v_A^2}{1-v_A^2}\right)$$

$$= m$$

$$P'_A = \gamma_A(\gamma_A m v_A) - \gamma_A v_A(\gamma_A m)$$

$$= 0$$

$$E'_B = \gamma_A(m \gamma_A) - v_A \gamma_A(-v_A \gamma_A m)$$

$$= \gamma_A^2 m + \gamma_A^2 v_A^2 m$$

$$= m(\gamma_A^2 + \gamma_A^2 v_A^2)$$

$$= m\left(\frac{1}{1-v_A^2} + \frac{v_A^2}{1-v_A^2}\right)$$

$$= m\left(\frac{1+v_A^2}{1-v_A^2}\right) = m \gamma'_B$$

(denne siste er litt strenger):

$$\gamma'_B = \frac{1}{\sqrt{1-v_B'^2}} = \frac{1}{\left(1 - \left(\frac{-2v_A}{1+v_A^2}\right)^2\right)^{\frac{1}{2}}}$$

$$= \frac{1}{\left(1 - \frac{4v_A^2}{(1+v_A^2)^2}\right)^{\frac{1}{2}}} = \frac{1}{\left(\frac{1+2v_A^2+v_A^4-4v_A^2}{(1+v_A^2)^2}\right)^{\frac{1}{2}}}$$

$$= \frac{1}{\left(\frac{(1-v_A^2)^2}{(1+v_A^2)^2}\right)^{\frac{1}{2}}} = \frac{1+v_A^2}{1-v_A^2}$$

Til slutt

$$P'_B = \gamma_A(\gamma_A m v_A) - \gamma_A v_A(\gamma_A m)$$

$$= -\gamma_A^2 m v_A - \gamma_A^2 m v_A$$

$$= -2\gamma_A^2 v_A m = -2v_A m \frac{1}{1-v_A^2}$$

$$= -2v_A m \frac{\gamma_B}{1+v_A^2} \quad (\text{Fra } \gamma'_B)$$

Oppsummering:

$$E'_A = m, \quad P'_A = 0$$

$$E'_B = m\left(\frac{1+v_A^2}{1-v_A^2}\right), \quad P'_B = \frac{-2v_A m}{1-v_A^2}$$

8)

$$E = \frac{E_{TOT}}{N}$$

$$E_{SI} = E \cdot C^2 = \frac{E_{TOT} \cdot C^2}{N}$$

Fra

$$\downarrow \quad f = \frac{E_{SI}}{h} = \frac{E_{TOT} \cdot C^2}{N \cdot h}$$

$$E_A = E_B = \gamma_A m$$

$$E_{TOT} = E_A + E_B = 2\gamma_A m$$

$$f = \frac{2mC^2}{(1-v_A^2)^{\frac{1}{2}} N \cdot h}$$

$$\lambda = \frac{c}{f}$$

Vi setter inn tall:

$$\lambda = 587.000 \text{ nm}$$

Wikipedia sier at dette er gult lys, som sammen med Mcast.

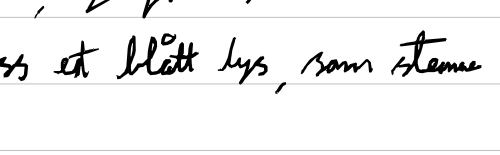
10)

$$E' = \gamma_{rel} E - v_{rel} \gamma_{rel} P$$

$$E'_x = \gamma_{rel} E_x \pm v_{rel} \gamma_{rel} E_y$$

$$= E_x \gamma_{rel} (1 \pm v_{rel})$$

Positiv betegn gir - i formel, negativ betegn gir + i formel.



Vi i tillegg for $v_{rel} P \gamma_{rel}$, som gir oss + når $P < 0$.

12) Se kurven av dekket i

13)

Vi setter inn, og får $\lambda' = 472.000 \text{ nm}$,

Som gir oss et blått lys, som stemmer med Mcast.

14)

$$f(v) = \sqrt{\frac{1+v}{1-v}}$$

Vi tar der-derivert rundt $v=0$

$$f(v, 0) = f(0) + \frac{f'(0)}{1!} (v)$$

Vi finner $f'(v)$ og $f''(v)$:

$$f'(v) = \frac{1}{2} \left(\frac{1+v}{1-v}\right)^{-\frac{1}{2}} \cdot \frac{1-v - (-1-v)}{(1-v)^2}$$

$$= \frac{\sqrt{1-v}}{\sqrt{1+v}}$$

$$f'(0) = \frac{\sqrt{1}}{\sqrt{1}} = 1. \quad \text{vi får:}$$

$$f(v) = 1 + v, \quad \text{Som gir oss}$$

$$\frac{\Delta \lambda}{\lambda} = (1 + v - 1)$$

$$\downarrow$$

$$\frac{\Delta \lambda}{\lambda} = v$$

Som er doppler-forskyning i relativistiske enheter.

$$\frac{\Delta \lambda}{\lambda} = \frac{v_{rel}}{c}$$

12) $\frac{\Delta \lambda}{\lambda} = \frac{\lambda' - \lambda}{\lambda} = \frac{\frac{hc}{E'} - \frac{hc}{E}}{\frac{hc}{E}} = \frac{E}{E'} - 1 = \frac{E}{E \gamma(1-v)} - 1$

har $-v$ fordi vi bruker $|v|$.

$$= \frac{\sqrt{1-v^2}}{(1-v)} - 1 = \frac{\sqrt{(1-v)(1+v)}}{(1-v)} - 1 = \left(\frac{\sqrt{1+v}}{\sqrt{1-v}} - 1\right)$$